

Detector-Conditioned Observability of Interference Under Coarse-Grained Measurement

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Abstract

Interference visibility is commonly treated as an intrinsic property of wave superposition. In practical systems, however, visibility is inferred through detector sampling and noise-affected measurements. We present a computational measurement framework integrating wave propagation, detector discretization, noise modeling, and visibility estimation. We find an empirical scaling: $B_{\min} \propto d^{-0.77}$, deviating from ideal Nyquist scaling due to a resolution–noise trade-off.

1. Introduction

Interference is a fundamental signature of coherence in wave systems. Real measurement systems introduce finite resolution, shot noise, and read noise, affecting observed visibility.

2. Measurement Stack Foundation

Pipeline: field $\rightarrow$ propagation $\rightarrow$ detector $\rightarrow$ noise $\rightarrow$ measurement. Includes Fresnel propagation, detector sampling, and noise modeling.

3. Extended Framework

Updated pipeline: field $\rightarrow$ propagation $\rightarrow$ detector $\rightarrow$ binning $\rightarrow$ noise $\rightarrow$ visibility $\rightarrow$ decision. Criterion: $V \geq 0.75$.

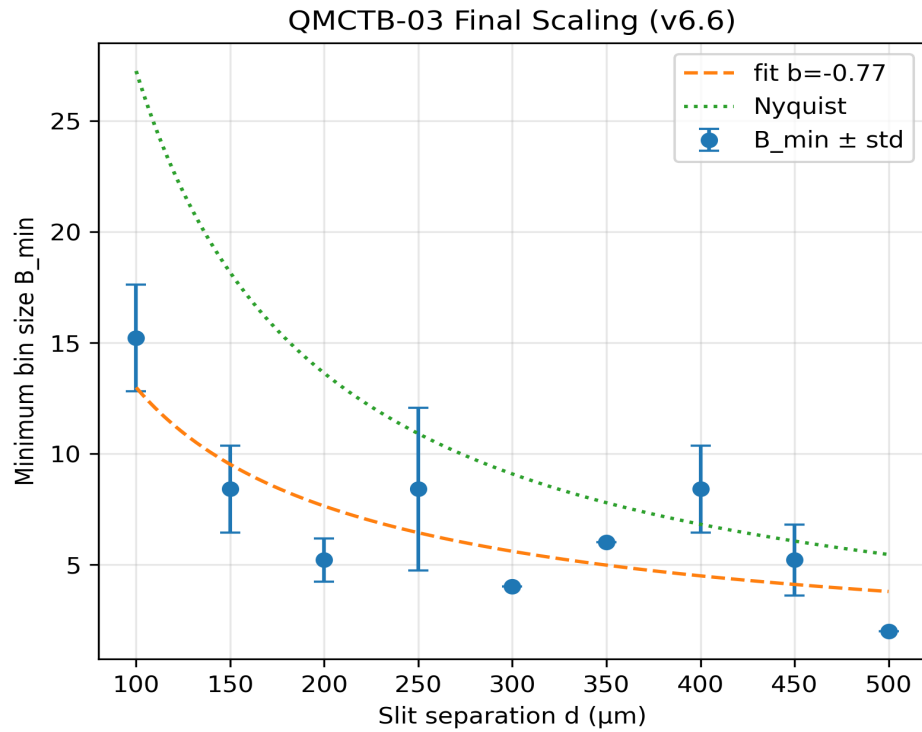
4. Experiment

Double-slit simulation with $d \in [100, 500] \mu\text{m}$. Outputs include mean $B_{\min}$ and standard deviation.

5. Results

We observe scaling: $B_{\min} \propto d^{-0.77}$.

Figure 1: Scaling of Detector Resolution vs Slit Separation



Minimum detector bin size ($B_{\min}$) vs slit separation (d). Dashed: fitted scaling. Dotted: Nyquist prediction.

6. Interpretation

High resolution preserves fringes but amplifies noise. Low resolution suppresses noise but blurs fringes, creating a trade-off.

7. Core Insight

Interference visibility is measurement-conditioned, not purely intrinsic.

8. Reproducibility

Code: <https://github.com/srikarr20/quantum-measurement-stack-demo>

9. Conclusion

Interference observability is probabilistic under noise and depends on detector characteristics.